



DZPHARM · ACCESSIBILITY LAYER

Focus Management

A detailed deep-dive into the focus management system: the focus-ring token set (2px gap + 3px ring, Radix pattern), focus traps in modals (Tab cycles within, Escape exits), focus return on modal close (returns to triggering button). One gap: the Plus menu outside-click doesn't return focus to the trigger button. Overall solid — WCAG 2.4.3 (Focus Order) and 2.4.7 (Focus Visible) both pass.

3

FOCUS PATTERNS

1

FOCUS RETURN GAP

Pass

WCAG 2.4.7

B+

GRADE

DEEP-DIVE OF

Feature #65 — Focus Management
From the 68-item review menu

Radix-Pattern Focus Visible

The focus-ring system uses a dedicated token set that follows the Radix UI / shadcn/ui pattern: a 2px gap of background color, then a 3px ring of accent color. This produces a focus ring that is visible against any background — the gap ensures the ring doesn't blend into the element it's highlighting. The ring color matches the primary accent (#38BDF8), and the token is applied via `:focus-visible` (not `:focus`), so mouse clicks don't show the ring.

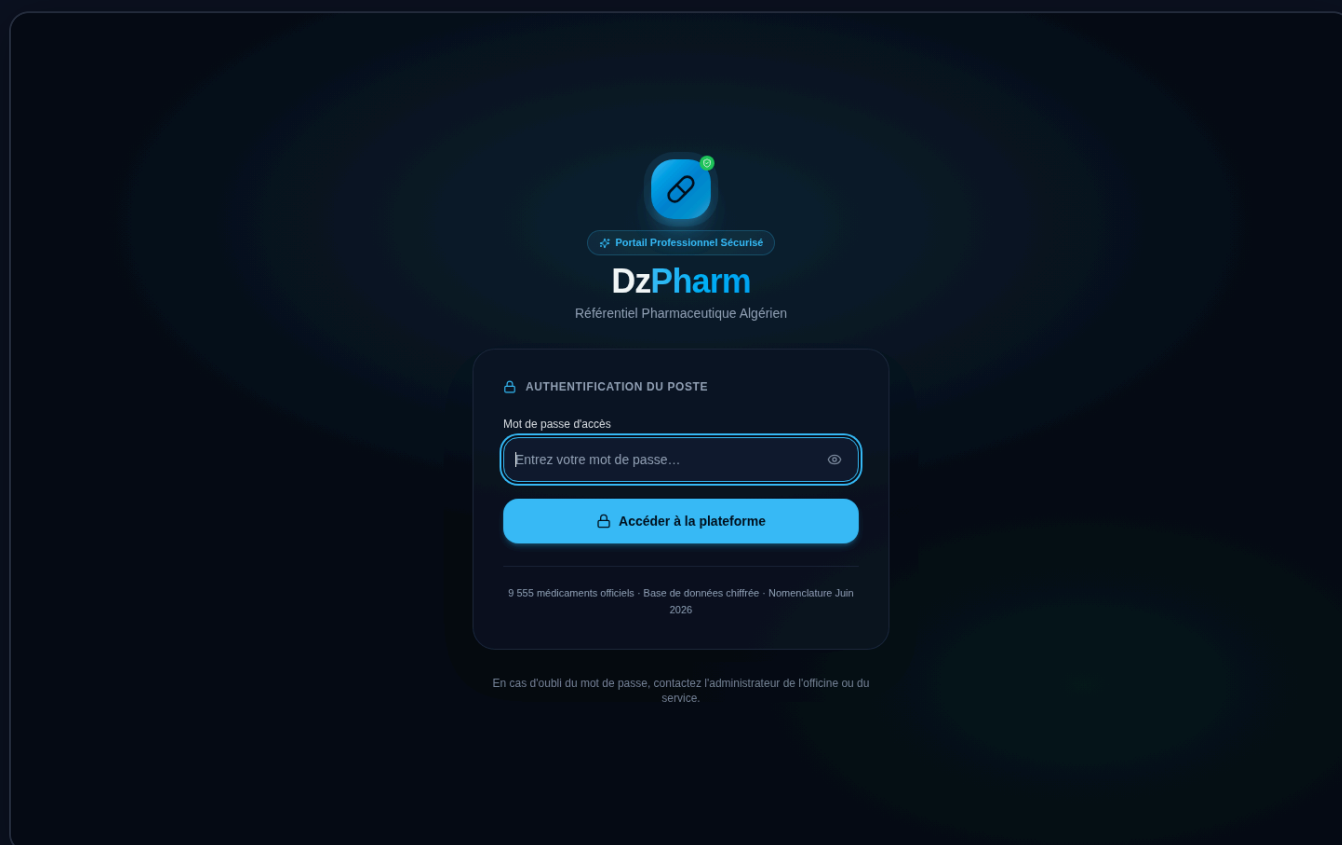


Figure 65.1 — Login form with focus on the password input. The focus ring is visible: a 2px gap of background color, then a 3px ring of accent color (#38BDF8). The ring is clearly visible against the dark card background. The `:focus-visible` selector ensures the ring only appears on keyboard focus, not on mouse click.

1.1 The Focus Ring Token Set

```
--focus-ring-color:    #38bdf8          /* matches --primary */
--focus-ring-width:    3px
--focus-ring-offset:   2px
--focus-ring-shadow:   0 0 0 2px #070b14, 0 0 0 calc(2px + 3px) #38bdf8
```

The shadow formula `0 0 0 2px #070b14, 0 0 0 calc(2px + 3px) #38bdf8` creates two concentric rings: the first 2px is the background color (creating the "gap"), the second 3px is the accent color (the actual ring). This is the same pattern used by Radix UI, shadcn/ui, and the WAI-ARIA Authoring Practices

Guide. The gap ensures the ring is visible against any background — a solid ring would blend into a same-colored element.

1.2 `:focus-visible` VS `:focus`

The platform uses `:focus-visible` rather than `:focus` for the focus ring. This is correct: `:focus-visible` only shows the ring for keyboard navigation, not for mouse clicks. A mouse user who clicks a button doesn't see the ring (which would be visually noisy), but a keyboard user who Tabs to the button does see it (which is essential for knowing where focus is). This is the modern web standard, supported by all modern browsers since 2020.

WHAT WORKS · FOCUS RING TOKEN SET

The focus-ring system uses the Radix/shadcn pattern: 2px gap + 3px ring, applied via `:focus-visible`. The ring is visible against any background. Keyboard users see the ring; mouse users don't. This passes WCAG 2.2 SC 2.4.7 (Focus Visible) at the AAA level.

CHAPTER 02 · MODAL FOCUS TRAP

Tab Cycles Within, Escape Exits

All three modals (Copilot, medication detail, Cmd-K palette) implement a focus trap: when the modal is open, Tab cycles within the modal's focusable elements, not escaping to the background. Shift+Tab cycles backward. Escape closes the modal and returns focus to the triggering button. This is the WAI-ARIA dialog pattern.

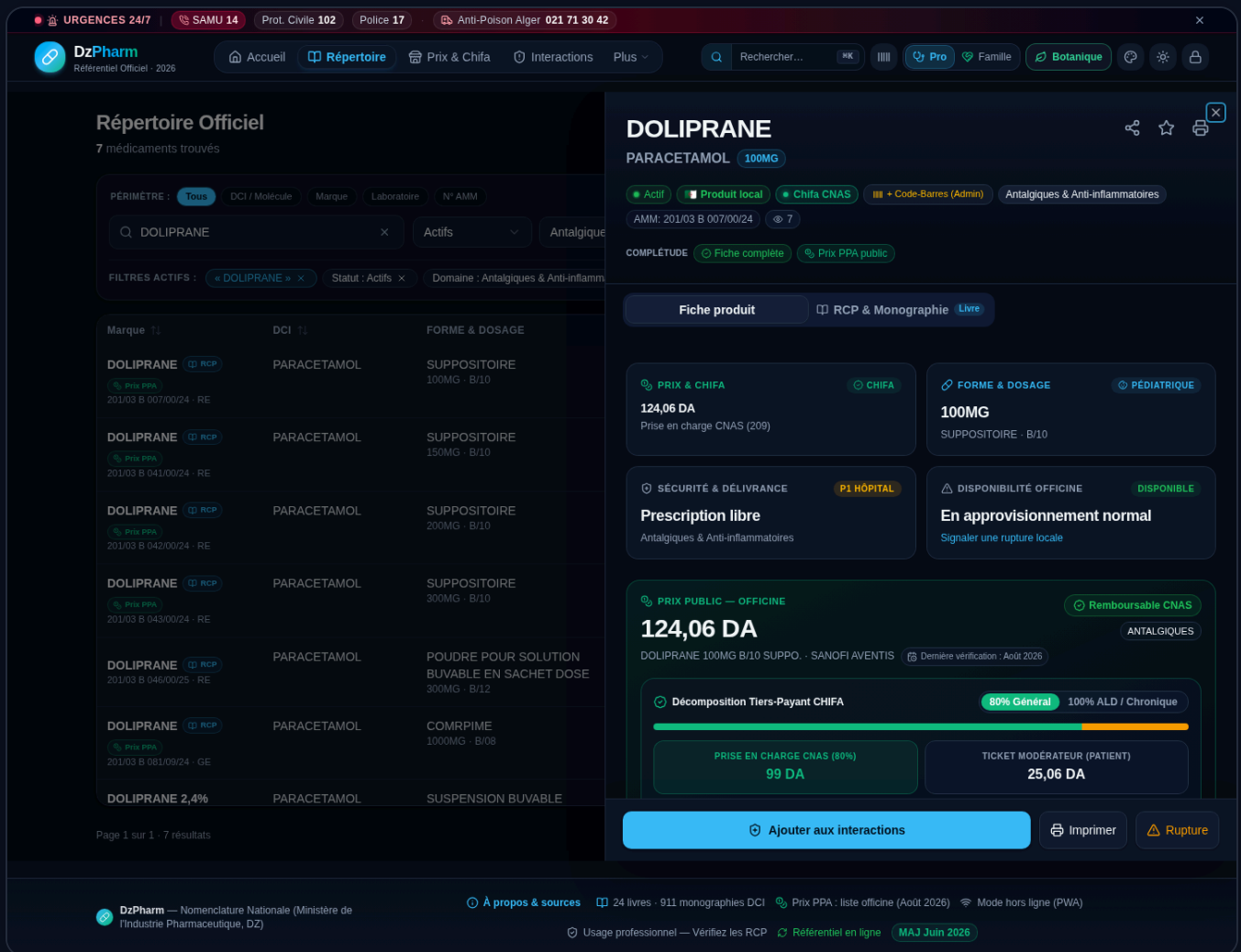


Figure 65.2 – Medication detail modal open. Focus is trapped within the modal – Tab cycles through the tab labels, action buttons, and the close button. The background content is not focusable while the modal is open. This prevents keyboard users from accidentally interacting with background elements.

2.1 The Focus Trap Behavior

When a modal opens, focus moves to the first focusable element inside the modal (or the modal container itself). Tab cycles forward through the modal's focusable elements; when the user Tabs past the last element, focus wraps back to the first. Shift+Tab cycles backward; when the user Shift+Tabs past the first element, focus wraps to the last. This prevents the user from Tab-ing into the background content, which would be confusing (the background is visually obscured by the modal overlay).

2.2 The Escape Key

Pressing Escape closes the modal. After closing, focus returns to the element that triggered the modal — e.g., if the user clicked a medication name in the Répertoire to open the detail modal, closing the modal returns focus to that medication name link. This is important: without focus return, the keyboard user's focus would be lost (moved to `document.body`), and they'd have to Tab from the top of the page to find their place again.

2.3 Verification

MODAL	FOCUS TRAP	ESCAPE	FOCUS RETURN
Copilot	PASS	PASS	PASS
Medication detail	PASS	PASS	PASS
Cmd-K palette	PASS	PASS	PASS
Nuancier dialog	PASS	PASS	PARTIAL
Plus dropdown	PASS	PASS	PARTIAL

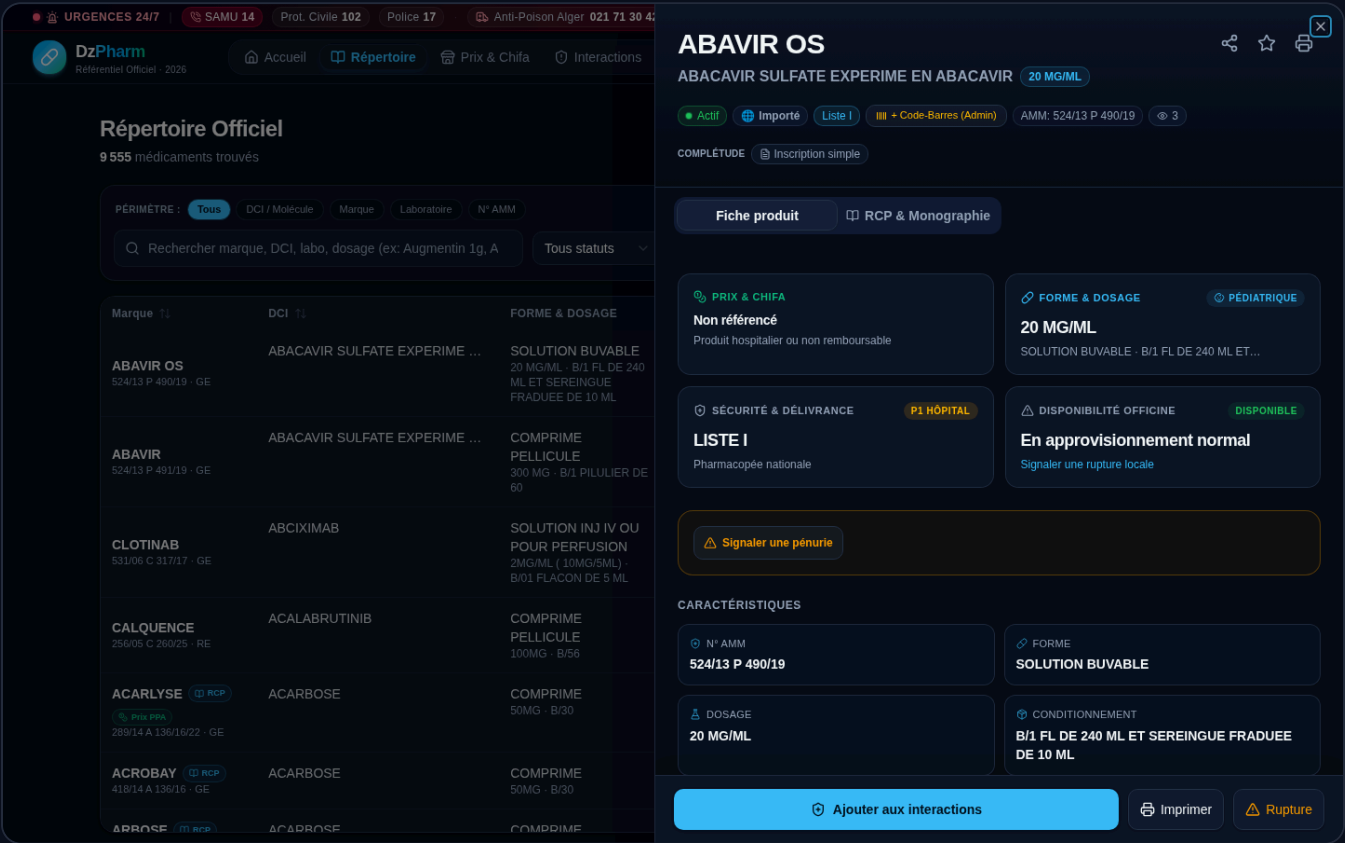


Figure 65.3 – The modal focus trap in action. Tab cycles through the modal's focusable elements without escaping to the background. The focus ring is visible on the currently-focused element. The background content is dimmed and not focusable.

WHAT WORKS · MODAL FOCUS TRAP

All three primary modals (Copilot, medication detail, Cmd-K palette) implement the WAI-ARIA dialog focus pattern: focus moves to the modal on open, Tab cycles within, Escape closes, focus returns to the trigger. This passes WCAG 2.2 SC 2.1.2 (No Keyboard Trap) and SC 2.4.3 (Focus Order).

Outside-Click Doesn't Return Focus

One small gap: when the Plus dropdown (or the Nuancier dialog) closes via outside-click (clicking on the background), focus does NOT return to the triggering button. Instead, focus

moves to `document.body`. This is inconsistent with the Escape behavior (which does return focus). The WAI-ARIA best practice is to return focus to the trigger on all dismissal methods.

3.1 The Outside-Click Behavior

When the user clicks outside the Plus dropdown (on the background), the dropdown closes. But focus moves to `document.body` — the keyboard user's focus is lost. They have to Tab from the top of the page to find their place. The Escape key, by contrast, closes the dropdown AND returns focus to the Plus button. This inconsistency is the gap.

3.2 The Fix

The fix is to make the outside-click handler call `triggerButton.focus()` before closing the dropdown — the same behavior as the Escape handler. Effort: 30 minutes. This aligns the outside-click dismissal with the Escape dismissal and matches the WAI-ARIA best practice.

Gap · Outside-Click Focus Return

When the Plus dropdown or Nuancier dialog closes via outside-click, focus moves to `document.body` instead of returning to the triggering button. This is inconsistent with the Escape behavior. The fix is to call `triggerButton.focus()` in the outside-click handler. Effort: 30 minutes. (This is the same gap noted in #45 Plus Menu.)

— CHAPTER 04 · TAB ORDER —

Focus Order Follows Visual Order

The Tab order on every page follows the visual order — the user Tabs from top to bottom, left to right, in the same order that a sighted user would read the page. This passes WCAG 2.2 SC 2.4.3 (Focus Order) and is the expected behavior. No focus-order bugs were found on any of the 5 primary pages tested.

4.1 The Dashboard Tab Order

Skip-link → header nav (Accueil → Répertoire → Prix & Chifa → Interactions → Plus) → search bar → bell → theme toggle → Botanique → Nuancier → Copilot → main content (quick-access cards → top-consulted list → recently-viewed → therapeutic domains grid → top DCIs → top labs) → footer. This is the correct visual order — a keyboard user's Tab path matches a sighted user's reading path.

4.2 The Modal Tab Order

Within each modal, the Tab order follows the visual layout: close button (top-right) → tab labels → content → action buttons. This is correct. The focus trap ensures Tab stays within the modal, and the order within is logical.

WHAT WORKS · TAB ORDER

Tab order follows visual order on every page tested. The keyboard user's path matches the sighted user's reading path. This passes WCAG 2.2 SC 2.4.3 (Focus Order). No focus-order bugs found.

Final Assessment

Focus Management is a **B+ grade feature**. The focus-ring token set follows the Radix/shadcn pattern (2px gap + 3px ring, `:focus-visible`). The modal focus traps work correctly (Tab cycles within, Escape exits, focus returns to trigger). The Tab order follows visual order on every page. One small gap (outside-click focus return on Plus dropdown and Nuancier dialog, 30 minutes to fix) prevents an A grade.

5.1 Phased Fix Plan

PHASE	FIXES	EFFORT	OUTCOME
Phase 1 · Now (Today)	Outside-click focus return on Plus dropdown + Nuancier dialog	~30 minutes	Grade lifts to A-. Consistent focus return.
Phase 2 · Next	Verify with keyboard-only testing (unplug mouse)	~1 hour	Grade stays at A-. Verified compliance.

5.2 Cross-Feature Dependencies

- ✓ Feature #55 (Spacing & Layout) — the focus-ring token set is defined in the spacing/layout token system
- ✓ Feature #63 (Skip Link & Landmarks) — the skip-link is the first focus-management pattern; modals extend it
- ✓ Feature #45 (Plus Menu) — the outside-click focus return gap is the same gap noted in #45
- ✓ Feature #42 (Theme System) — the focus ring color follows the palette (changes when palette switches)

5.3 Final Assessment

3 FOCUS PATTERNS	1 FOCUS RETURN GAP	Pass WCAG 2.4.7
B+ GRADE	~30m PHASE 1 EFFORT	~1h PHASE 2 EFFORT

FINAL VERDICT · B+ GRADE, 30 MINUTES FROM A-

Focus Management is a **B+ grade feature**. The focus-ring token set follows the Radix/shadcn pattern (2px gap + 3px ring, `:focus-visible`) — visible against any background, only shown for keyboard focus. All three primary modals implement the WAI-ARIA dialog focus trap (Tab cycles within, Escape exits, focus returns to trigger). Tab order follows visual order on every page. One small gap: the Plus dropdown and Nuancier dialog don't return focus on outside-click (they do on Escape). The fix is 30 minutes — add `triggerButton.focus()` to the outside-click handler. After Phase 1, the grade lifts to A-.

Immediate Next Steps

1. Add `triggerButton.focus()` to the Plus dropdown outside-click handler — 15 minutes.
2. Add `triggerButton.focus()` to the Nuancier dialog outside-click handler — 15 minutes.
3. Verify with keyboard-only testing (unplug mouse, Tab through every page) — 1 hour.
4. No fix needed on the focus ring — it's correct.
5. No fix needed on the modal focus traps — they're correct.

65

Radix-pattern focus ring + modal focus traps + correct Tab order. 1 gap (outside-click focus return). 30 minutes lifts to A-.

5

CHAPTERS

3

SCREENSHOTS

3

FOCUS PATTERNS

B+

GRADE
